

Chapter 3

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May 13, 2024

Definition. Let $\mathcal{T}$ be a topology on X and let $\mathcal{B} \subset \mathcal{T}$. Then $\mathcal{B}$ is a **basis** for the topology $\mathcal{T}$ if and only if every open set in $\mathcal{T}$ is a union of elements in $\mathcal{B}$. If $B \in \mathcal{B}$ we say B is a **basis element** or a **basis open set**. ■

Claim 3.1. Let $(X, \mathcal{T})$ be a topological space, and let $\mathcal{B}$ be a collection of subsets of X . Then $\mathcal{B}$ is a basis of $\mathcal{T}$ if and only if

1. $\mathcal{B} \subseteq \mathcal{T}$
2. $\forall U \in \mathcal{T}$ and $\forall p \in U$, $\exists V \in \mathcal{B} \ni p \in V \subseteq U$

Proof. $\implies$: Suppose $\mathcal{B}$ is a basis of $\mathcal{T}$. Then by definition $\mathcal{B} \subseteq \mathcal{T}$. Moreover, by definition of basis, $\forall U \in \mathcal{T}$, $\exists \{B_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{B}$ a subcollection such that $U = \bigcup_{\alpha \in \lambda} B_\alpha$. Thus by definition of union, $\forall p \in U$, $\exists B_\alpha \ni p \in B_\alpha \subseteq U$.

$\impliedby$: Suppose 1 and 2 are true. Then $\forall U \in \mathcal{T}$, $\forall p \in U$ $\exists V_p \in \mathcal{B} \ni p \in V_p \subseteq U$ then $U = \bigcup_{p \in U} V_p$. Then by definition of a basis of topological space, $\mathcal{B}$ is a basis of $\mathcal{T}$. □

Problem 3.2a. Let $\mathcal{B}_1 = \{(a, b) \subset \mathbb{R} \mid a, b \in \mathbb{Q}\}$. Show that $\mathcal{B}_1$ is a basis for the standard topology on $\mathbb{R}$.

Solution. $\mathcal{B}_1 \subseteq \mathcal{T}$. $\forall U \in \mathcal{T}$, $\forall p \in U$, $\exists \epsilon \in \mathbb{R}^+ \ni p \in (p - \epsilon, p + \epsilon) \subseteq U$. By the density of rationals in $\mathbb{R}$, $\exists r \in \mathbb{Q} \ni p - \epsilon \leq r < p$ and $\exists s \in \mathbb{Q} \ni p < s \leq p + \epsilon$, thus $p \in (r, s) \subseteq (p - \epsilon, p + \epsilon) \subseteq U$ and $(r, s) \in \mathcal{B}_1$. Thus by claim 3.1, $\mathcal{B}_1$ is a basis of $\mathcal{T}$. ■

Problem 3.2b. Let $\mathcal{B}_2 = \{(a, b) \cup (c, d) \subseteq \mathbb{R} \mid a, b, c, d \in \mathbb{R} \setminus \mathbb{Q}, a < b < c < d\}$. Show $\mathcal{B}_2$ is a basis for the standard topology on $\mathbb{R}$.

Solution. $\mathcal{B}_2 \subseteq \mathcal{T}$. $\forall U \in \mathcal{T}$, $\forall p \in U$, $\exists \epsilon \in \mathbb{R}^+ \ni p \in (p - \epsilon, p + \epsilon) \subseteq U$. By the density of irrationals in $\mathbb{R}$, $\exists a, b, c, d \in \mathbb{R} \setminus \mathbb{Q} \ni p - \epsilon < a < p < p + \frac{\epsilon}{4} < c < p + \frac{3\epsilon}{4} < d < p + \epsilon$. Thus $p \in (a, b)$ and $(a, b) \cup (c, d) \subseteq (p - \epsilon, p + \epsilon) \subseteq U$. Thus $\mathcal{B}_2$ is a basis by claim 3.1. ■

Claim 3.3. Suppose X is a set and $\mathcal{B}$ is a collection of subsets of X . Then $\mathcal{B}$ is a basis for some topology on X iff

1. each point of X is in some element of $\mathcal{B}$
2. if $U, V \in \mathcal{B}$ and $p \in U \cap V$ then $\exists W \in \mathcal{B} \ni p \in W \subseteq U \cap V$.

Proof. $\implies$: Suppose $\mathcal{B}$ is a basis for some topology $\mathcal{T}$. Since $X \in \mathcal{T}$, by claim 3.1 $\forall x \in X, \exists V \in \mathcal{B} \ni p \in V \subseteq X$. Thus 1 is true. $\forall U, V \in \mathcal{B}$, $U \cap V$ is a open set of X . By claim 3.1, $\forall p \in U \cap V, \exists W \in \mathcal{B} \ni p \in W \subseteq U \cap V$. Thus 2 is true.

$\impliedby$: Suppose 1 and 2 are true. Let $\mathcal{T}$ be the collections of sets formed by taking unions of collections of elements of $\mathcal{B}$. $\emptyset \in \mathcal{T}$ take the union of a empty collection. Since $\forall x \in X, \exists B_x \in \mathcal{B} \ni x \in B_x \implies X = \bigcup_{x \in X} B_x \implies X \in \mathcal{T}$. Let $\{U_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{T}$, by definition $\forall \alpha \exists \{B_{\alpha, \beta}\}_\beta \subseteq \mathcal{B} \ni U_\alpha = \bigcup_\beta B_{\alpha, \beta} \implies \bigcup_\alpha U_\alpha = \bigcup_\alpha \bigcup_\beta B_{\alpha, \beta} \in \mathcal{T}$. Since $\forall B_1, B_2 \in \mathcal{B}, \forall p \in B_1 \cap B_2, \exists B_3 \in \mathcal{B} \ni p \in B_3 \subseteq B_1 \cap B_2 \implies B_1 \cap B_2 \in \mathcal{T}$, by taking the union of all B_3 per element of $p \in B_1 \cap B_2$. Thus $\forall U, V \in \mathcal{T}, \exists \{U_\alpha\}_\alpha, \{V_\beta\}_\beta \subseteq \mathcal{B}$ such that $U = \bigcup_\alpha U_\alpha$ and $V = \bigcup_\beta V_\beta$. Thus $U \cap V = \bigcup_\alpha U_\alpha \cap \bigcup_\beta V_\beta = \bigcup_{\alpha, \beta} U_\alpha \cap V_\beta \in \mathcal{T}$ since $\{U_\alpha \cap V_\beta\}_{\alpha, \beta} \subseteq \mathcal{T}$. Thus $\mathcal{T}$ is a topology and $\mathcal{B}$ is a basis of a topology. $\square$

Definition. $\mathbb{R}_{LL}$ is the lower limit topology on $\mathbb{R}$ which is generated by the basis $\{[a, b) \mid a, b \in \mathbb{R}, a < b\}$. $\blacksquare$

Problem 3.4. Check the basis provided for $\mathbb{R}_{LL}$ is in fact a basis

Solution. $\forall x \in \mathbb{R}, x \in [x, x+1) \in \mathcal{B}$. $\forall [p, q), [r, s) \in \mathcal{B}$, WLOG $r < q$ and $s > p$, because otherwise $[p, q) \cap [r, s) = \emptyset$. Thus $[p, q) \cap [r, s) = [\max(r, s), \min(q, s)) \in \mathcal{B}$. Thus $\mathcal{B}$ is a basis. $\blacksquare$

Claim 3.5. Every open set in $\mathbb{R}_{std}$ is a open set in $\mathbb{R}_{LL}$ but not vice versa

Proof. It suffices to prove that $(a, b) \in \mathbb{R}_{LL}$ for $a < b$ since they form a basis of $\mathbb{R}_{std}$. $\forall p \in (a, b), p \in [p, b) \implies (a, b) = \bigcup_{p \in (a, b)} [p, b) \implies (a, b) \in \mathbb{R}_{LL}$. Thus $\mathbb{R}_{std} \subseteq \mathbb{R}_{LL}$.

Assume the contrary that $[a, b)$ is open in $\mathbb{R}_{std}$. By claim 2.3, $\exists \epsilon \in \mathbb{R}^+ \ni (a - \epsilon, a + \epsilon) \subseteq [a, b)$, but this is a contradiction. Thus $[a, b)$ is not open in $\mathbb{R}_{LL}$. $\square$

Definition. Suppose $\mathcal{T}$ and $\mathcal{T}'$ are two topologies on X . If $\mathcal{T} \subseteq \mathcal{T}'$ then $\mathcal{T}'$ is **finer** than $\mathcal{T}$. Alternatively $\mathcal{T}'$ is **coarser** than $\mathcal{T}$. Additionally, we say **strictly finer** or **strictly coarser** if $\mathcal{T} \neq \mathcal{T}'$. $\blacksquare$

Problem 3.6. Give a example of 2 topologies on $\mathbb{R}$ such that neither topologies are comparable.

Solution. $\mathbb{R}_{std}$ and $\mathbb{R}_{fin}$ or $\mathbb{R}_{count}$ $\blacksquare$

Definition. Construct the Double Headed Snake Topology. Let $\mathbb{R}_{+00}$ be the set $\mathbb{R}_+$ with points $0'$ and $0''$. Put a topology on it whose basis are all intervals (a, b) or $(0, b) \cup \{0'\}$ or $(0, b) \cup \{0''\}$ for $a, b \in \mathbb{R}_+$. $\blacksquare$

Problem 3.7. Show the collection of sets that we specify as a basis for the Double Headed Snake actually forms a basis

Solution. $\forall x \in \mathbb{R}_{+00}$, $x \in \mathbb{R}_+$ or $x = 0'$ or $x = 0''$. If $x \in \mathbb{R}_+$, $\forall \epsilon > 0$, $x \in (0, x + \epsilon)$. If $x = 0'$ then $x \in (0, 1) \cup \{0'\}$ and the same for $0''$.

$\forall U, V \in \mathcal{B}$ wlog U, V are the same type or they are simple open intervals.

1. U and V are the same type. WLOG they are both $0'$ as the case for $0''$ is symmetric and if they are simple open intervals it defaults to the case below. $U = (0, b) \cup \{0'\}$ and $V = (0, d) \cup \{0'\}$. $U \cap V = (0, \min(b, d)) \cup \{0\} \in \mathcal{B}$.
2. U and V are open intervals. $U = (a, b)$ and $V = (c, d)$ where $\max(a, c) < \min(b, d)$ otherwise the intersection is empty. $U \cap V = (\max(a, c), \min(b, d)) \in \mathcal{B}$

Thus $\mathcal{B}$ is a basis for the topology by 3.3. ■

Problem 3.8. *In the Double Headed Snake Topology every point is a closed set, and it is impossible to find disjoint open sets U and V such that $0' \in U$ and $0'' \in V$*

Proof. $\forall x \in \mathbb{R}_{+00}$. If $x \in \mathbb{R}_+$, then $\mathbb{R}_{+00} \setminus \{x\} = (0, x) \cup \{0', 0''\} = ((0, x) \cup \{0'\}) \cup ((0, x) \cup \{0''\}) \implies \mathbb{R}_{+00} \setminus \{x\} \in \mathcal{T} \implies \{x\}$ is closed by claim 2.14. If $x = 0'$, then $A = \mathbb{R}_{+00} \setminus \{0'\} = \mathbb{R}_+ \cup \{0''\} = \mathbb{R}_+ \cup (0, 1) \cup \{0'\} \implies A \in \mathcal{T} \implies \{0'\}$ is closed. The argument is symmetric for $0''$.

Assume the contrary $\exists U, V \in \mathcal{T} \ni U \cap V = \emptyset, 0' \in U, 0'' \in V$. U is of the form $I \cup (0, a) \cup \{0'\}$ and $J \cup (0, b) \cup \{0''\}$. I and J are union of open intervals of $\mathbb{R}_+$. Since $U \cap V = \emptyset \implies (I \cup (0, a)) \cap (J \cup (0, b)) = \emptyset \implies (0, a) \cap (0, b) = \emptyset$ which is not true if $0 < a$ and $0 < b$. Thus a contradiction occurs and it is impossible to find disjoint open sets U and V such that $0' \in U$ and $0'' \in V$. □

Subbases

Problem 3.13. *A basis for a topology is also a subbasis for that topology*

Solution. Let $\mathcal{B}_1$ be a basis for a given topology. Let $\mathcal{B}_2$ be the set collection consisting of the finite intersections of $\mathcal{B}_1$. Thus $\forall A \in \mathcal{B}_2, \exists \{U_i\}_{i=1}^n \subseteq \mathcal{B}_1 \ni A = \bigcap_{i=1}^n U_i$. Since $\mathcal{B}_1$ is a basis and thus $\forall x \in X, \exists U_x \in \mathcal{B}_1 \ni x \in U_x$ and since $U_x \in \mathcal{B}_2$, the first condition is trivially satisfied. $\forall A_1, A_2 \in \mathcal{B}_2, A_1 \cap A_2 = (\bigcap_{i=1}^n U_i) \cap (\bigcap_{i=1}^m V_i) \in \mathcal{B}_2$, by definition of $\mathcal{B}_2$. Thus by claim 3.3, $\mathcal{B}_2$ is a basis thus by definition of subbasis, $\mathcal{B}_1$ is a subbasis. ■

Problem 3.14. *Show that $\mathbb{R}$ with the standard topology has a subbasis S consisting of rays $(-\infty, a)$ and (a, ∞) for $a \in \mathbb{R}$.*

Solution. $\mathcal{B}_1$ consist of all finite intersections of S . We want to show that $\mathcal{B} := \{(a, b) \mid a, b \in \mathbb{R}, a < b\} \subseteq \mathcal{B}_1$. $\forall (a, b) \in \mathcal{B}, (a, b) = (a, \infty) \cap (-\infty, b)$. Thus $\mathcal{B} \subseteq \mathcal{B}_1$. Thus $\mathcal{B}_1$ is a basis. Thus S is a subbasis. ■

Claim 3.15. *Let $(X, \mathcal{T})$ be a topological space, and let S be a collection of subsets of X . Then S is a subbasis for $\mathcal{T}$ if and only if:*

1. $S \subseteq \mathcal{T}$
2. $\forall U \in \mathcal{T}$ and $\forall p \in U, \exists \{V_i\}_{i=1}^n \ni p \in \bigcap_{i=1}^n V_i \subseteq U$

Proof. $\implies$: Suppose S is a subbasis for $\mathcal{T}$. Let $\mathcal{B}$ be its corresponding basis. Since $\mathcal{B}$ is a basis, $\mathcal{B} \subseteq \mathcal{T}$. Since $S \subseteq \mathcal{B} \implies S \subseteq \mathcal{T}$. By 3.1, $\forall U \in \mathcal{T}$ and $\forall p \in U, \exists V \in \mathcal{B} \ni p \in V \subseteq U$. By definition V is a finite intersection of a collection of elements in S . Thus 2 is also satisfied.

$\impliedby$: Suppose 1 and 2 are satisfied. Let $\mathcal{B}$ be the collection of finite intersections of S . 3.1 is satisfied thus $\mathcal{B}$ is a basis. Thus S is a subbasis. □

Claim 3.16. *Suppose X is a set and S is a collection of subsets of X . Then S is a subbasis $\iff \forall x \in X, \exists U_x \in S \ni x \in U_x$*

Proof. $\implies$: Suppose S is a subbasis. Assume the theorem isn't true, $\exists x \in X \ni \forall U \in S, x \notin U$. Let $\mathcal{B}$ be its corresponding basis, ie collection of finite intersections. Thus x isn't in any element of $\mathcal{B}$. Thus $\mathcal{B}$ isn't a basis which is a contradiction. Thus the theorem is true.

$\impliedby$: Trivially accomplishes 3.3. □

Subspaces

Definition. Let $(X, \mathcal{T})$ be a topological space. For $Y \subseteq X$, the collection

$$\mathcal{T}_Y := \{U \mid U = V \cap Y \text{ for some } V \in \mathcal{T}\}$$

is a topology on Y called the subspace topology. ■

Claim 3.25. *Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$. Then the collection of sets $\mathcal{T}_Y$ is in fact a topology on Y*

Proof. $\emptyset \in \mathcal{T} \implies \emptyset \cap Y = \emptyset \in \mathcal{T}_Y$. $X \in \mathcal{T} \implies X \cap Y = Y \in \mathcal{T}_Y$. $\forall U_1, U_2 \in \mathcal{T}_Y, \exists V_1, V_2 \in \mathcal{T} \ni U_1 = V_1 \cap Y$ and $U_2 = V_2 \cap Y$. Thus $U_1 \cap U_2 = V_1 \cap Y \cap V_2 \cap Y = V_1 \cap V_2 \cap Y = (V_1 \cap V_2) \cap Y$. $V_1 \cap V_2 \in \mathcal{T} \implies (V_1 \cap V_2) \cap Y \in \mathcal{T}_Y \implies U_1 \cap U_2 \in \mathcal{T}_Y$. Let $\{U_\alpha\}_{\alpha \in \lambda}$ be collection in $\mathcal{T}_Y$. $U_\alpha = V_\alpha \cap Y$. Thus $\bigcup_{\alpha \in \lambda} U_\alpha = \bigcup_{\alpha \in \lambda} V_\alpha \cap Y = (\bigcup_{\alpha \in \lambda} V_\alpha) \cap Y \implies \bigcup_{\alpha \in \lambda} U_\alpha \in \mathcal{T}_Y$. Thus $\mathcal{T}_Y$ is a topology on Y . □

Problem 3.26. *Consider $Y = [0, 1)$ as a subspace of $\mathbb{R}_{std}$. Is $[\frac{1}{2}, 1)$ open, closed, both, or neither?*

Proof. $[0, \frac{1}{2}) = (-\frac{1}{2}, \frac{1}{2}) \cap Y \implies [0, \frac{1}{2}) \in \mathcal{T}_Y$. Thus $[\frac{1}{2}, 1)$ is closed. $(\frac{1}{2}, 1)$ is open. Assume the contrary, that $[\frac{1}{2}, 1)$ is open. Then $(\frac{1}{2}, 1) \cap [\frac{1}{2}, 1) = \{\frac{1}{2}\}$ is open which is a contradiction (look at $\mathbb{R}_{std}$ and intersect with Y). Thus $[\frac{1}{2}, 1)$ is not open. □

Claim 3.28. *Let $(Y, \mathcal{T}_Y)$ be a subspace of $(X, \mathcal{T})$. A subset $C \subseteq Y$ is closed in $(Y, \mathcal{T}_Y)$ if and only if $\exists D \subseteq X$, closed in $(X, \mathcal{T})$, such that $D \cap Y = C$*

Proof. $\implies$: Suppose $C \subseteq Y$ is closed in $(Y, \mathcal{T}_Y)$. By 2.14, $Y \setminus C$ is open. By definition of subspace, $\exists U \in \mathcal{T} \ni U \cap Y = Y \setminus C$. Thus $X \setminus U$ is closed. $(X \setminus U) \cap Y = Y \setminus (U \cap Y) = Y \setminus (Y \setminus C) = C$.

$\Leftarrow$: Suppose $\exists D \subseteq X$, closed, such that $D \cap Y = C$. Then $X \setminus D$ is an open set. Thus $(X \setminus D) \cap Y = Y \setminus (D \cap Y) = Y \setminus C$ is open in $\mathcal{T}_Y$. Thus C is closed in $\mathcal{T}_Y$. □

Claim 3.29. *Let $(Y, \mathcal{T}_Y)$ be a subspace of $(X, \mathcal{T}_X)$. A subset $C \subseteq Y$ is closed in $(Y, \mathcal{T}_Y)$ iff $Cl_X(C) \cap Y = C$.*

Proof. $\implies$: $C \subseteq Y$ is closed in $(Y, \mathcal{T}_Y)$. $C \subseteq Cl_X(C) \cap Y$. $\forall x \in Cl_X(C) \cap Y, \forall U \in \mathcal{T} \ni x \in U, U \cap C \neq \emptyset$. Since $x \in Y$ and $C \subseteq Y$, $(U \cap Y) \cap C \neq \emptyset$. Thus x is a limit point of C in $(Y, \mathcal{T}_Y)$. Thus $x \in Cl_Y(C) = C$. Thus $Cl_X(C) \cap Y = C$.

$\Leftarrow$: Suppose $Cl_X(C) \cap Y = C$. Then C is closed by 3.28 □

Claim 3.30. *Let $(X, \mathcal{T})$ be a topological space, and let $(Y, \mathcal{T})$ be a subspace. If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $\mathcal{B}_Y := \{B \cap Y \mid B \in \mathcal{B}\}$ is a basis for $\mathcal{T}_Y$.*

Proof. Since $\mathcal{B} \subseteq \mathcal{T} \implies \mathcal{B}_Y \subseteq \mathcal{T}_Y$. $\forall U_Y \in \mathcal{T}_Y \exists U \in \mathcal{T} \ni U \cap Y = U_Y$. $\forall p \in U, \exists V \in \mathcal{B} \ni p \in V \subseteq U$. $\forall p \in U_Y, \exists V \in \mathcal{B} \ni p \in V \subseteq U$, since $p \in Y, p \in V \cap Y \in \mathcal{B}_Y$. Moreover since $V \subseteq U \implies V \cap Y \subseteq U_Y$. Thus by 3.1, $\mathcal{B}_Y$ is a basis for $\mathcal{T}_Y$. □

Product Spaces

Problem 3.32. Verify the collection of basic open sets above satisfies theorem 3.3, thus confirming this collection is a basis for a topology

Solution. $X \in \mathcal{T}_X$ and $Y \in \mathcal{T}_Y$. Let $\mathcal{B} = \mathcal{T}_X \times \mathcal{T}_Y$. Thus $X \times Y \in \mathcal{B}$. Thus 1 is satisfied. $\forall A, B \in \mathcal{B}$, $A = U_1 \times V_1$ and $B = U_2 \times V_2$. $A \cap B = (U_1 \cap U_2) \times (V_1 \cap V_2) \in \mathcal{B}$ since $U_1 \cap U_2 \in \mathcal{T}_X$ and $V_1 \cap V_2 \in \mathcal{T}_Y$. Thus $\mathcal{B}$ is a basis. ■

Claim 3.35. Show that the product topology on $X \times Y$ is the same as the topology generated by the subbasis $S = \{\pi_x^{-1}(U) \mid U \in \mathcal{T}_X\} \cup \{\pi_y^{-1}(V) \mid V \in \mathcal{T}_Y\}$.

Proof. $\forall U \in \mathcal{T}_X$ and $\forall V \in \mathcal{T}_Y$, $\pi_x^{-1}(U) = U \times Y$ and $\pi_y^{-1}(V) = X \times V$, both of which are open since $X \in \mathcal{T}_X$ and $Y \in \mathcal{T}_Y$. Thus $S \subseteq \mathcal{T}$. $\forall U \in \mathcal{T}$, U is formed by the union of the products of open sets of X and Y . Thus $U = \bigcup_{\alpha \in \lambda} (U_\alpha \times V_\alpha)$ for $U_\alpha \in \mathcal{T}_X$ and $V_\alpha \in \mathcal{T}_Y$. Thus $\forall p \in U$, $\exists \alpha \in \lambda \ni p \in U_\alpha \times V_\alpha$. $U_\alpha \times V_\alpha = U_\alpha \times Y \cap X \times V_\alpha = \pi_x^{-1}(U_\alpha) \cap \pi_y^{-1}(V_\alpha)$. Thus $p \in \pi_x^{-1}(U_\alpha) \cap \pi_y^{-1}(V_\alpha) \subseteq U$. Thus S is a subbasis of $\mathcal{T}$ the product topology on $X \times Y$ □

Claim 3.37. A basis for the product topology on $\prod_{\alpha \in \lambda} X_\alpha$ is a collection of all sets of the form $\prod_{\alpha \in \lambda} U_\alpha$ where U_α is open in X_α and $U_\alpha = X_\alpha$ for all but finitely many α .

Proof. Let $\mathcal{T}$ be the product topology on $\prod_{\alpha \in \lambda} X_\alpha$ and $\mathcal{B}$ be the collection described above. $\forall U \in \mathcal{B}$, by definition $U = \prod_{\alpha \in \lambda} U_\alpha$ and U_α is open in X_α , by definition of product topology $U \in \mathcal{T}$. Thus $\mathcal{B} \subseteq \mathcal{T}$. $\forall U \in \mathcal{T}$, $U = \bigcup_{\beta \in \varphi} \prod_{\alpha \in \lambda} U_{\beta, \alpha}$. Thus $\forall p \in U$, $\exists \beta_p \in \varphi \ni p \in \prod_{\alpha \in \lambda} U_{\beta_p, \alpha} \in \mathcal{B}$. By definition of union, $\prod_{\alpha \in \lambda} U_{\beta_p, \alpha} \subseteq U$. Thus $\forall U \in \mathcal{T}$, $\forall p \in U$, $\exists V \in \mathcal{B} \ni p \in V \subseteq U$. Thus by claim 3.1, $\mathcal{B}$ is a basis on the product topology. □

Problem 3.38a. Let T be a topology on 2^X with a basis generated by subbasis $S := \{U(a, \delta) \mid a \in X, \delta \in \{0, 1\}\}$ where $U(a, \delta) := \{f \in 2^X \mid f(a) = \delta\}$. Show that every basic open set in 2^X is both open and closed.

Solution. Let $\mathcal{B}$ be the basis generated by S . $\forall A \in \mathcal{B}$, A is by definition open.

$\forall a \in X$, $2^X - U(a, 1) = U(a, 0)$. Thus $\forall A \in \mathcal{B}$, A is both open and closed. Thus $\forall B \in \mathcal{B}$, B is closed since $B = \bigcap_{\alpha \in \lambda} A_\alpha$ thus B is closed. ■

Problem 3.38b. Show that if a collection of subbasic open sets of 2^X has the property that every point of 2^X lies in at least one of those subbasic open sets, then there are two subbasic open sets in that collection such that every point of 2^X lies in one of those two subbasic sets.

Solution. ■